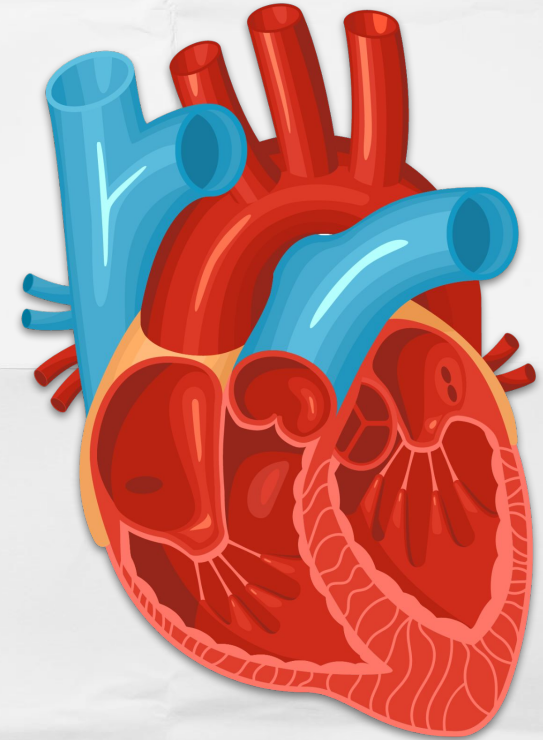


Automated Geometric Stenosis Quantification and CAD-RADS Support for Coronary Artery Disease Assessment and Patient Prioritization

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Mathematical Engineering in Data Science

PhySense Update 3



Problems at Hospital de la Santa Creu i Sant Pau

1. Elevated Reporting Times

- 30 minutes for non pathological cases.
- 1-2 hours for more complex cases.

2. Fragmentation of the Workflow

3. Lack of Automatization

4. No Patient Prioritization

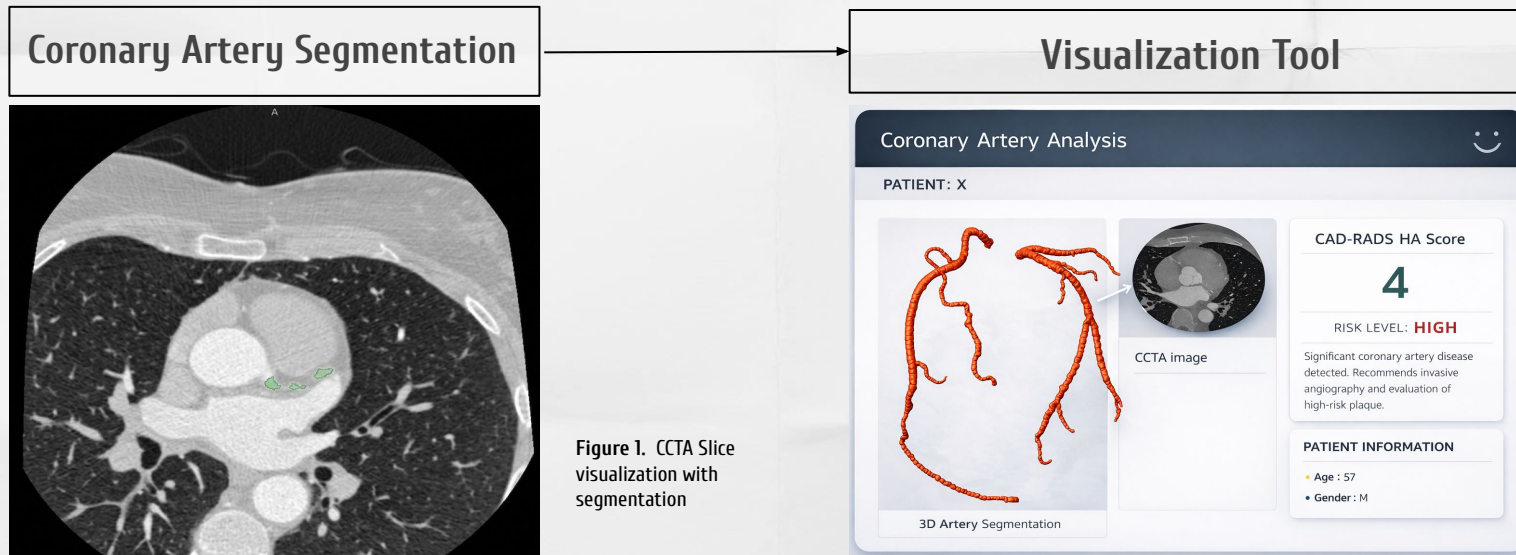
The hospital requires a shift from manual, time-consuming assessment to an automated AI pipeline that handles:

IMAGE ANALYSIS → QUANTIFICATION → REPORTING → PRIORITIZATION

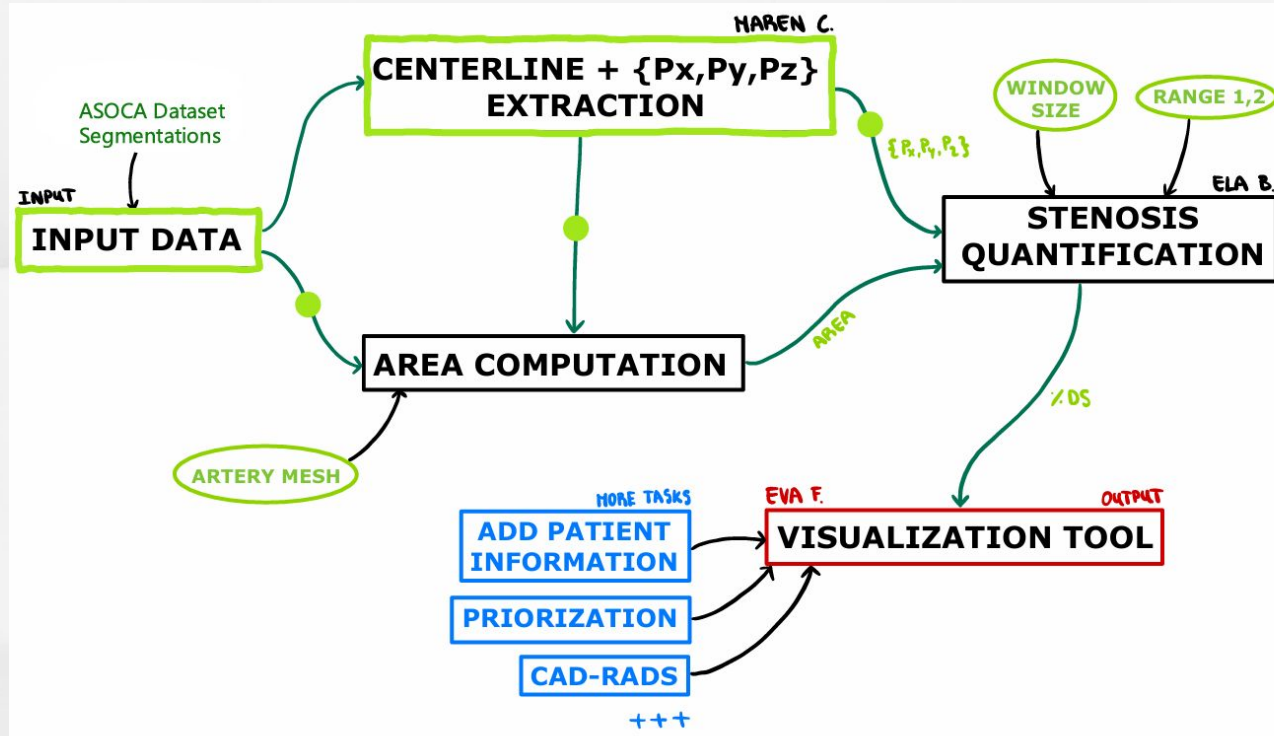
Project Goals & Scope

Main Mission

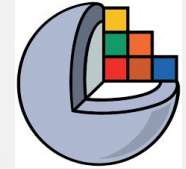
Make a bridge between **raw geometric data** and **clinical decision-making** by developing an automated support pipeline for quantification, patient prioritization and CAD diagnosis. (Acebes, 2023 [1])



Proposed Workflow: Current State



The Input Data: ASOCA Annotations



CCTA "Volume": Normal_1.nrrd

Annotations "Segmentation": Normal_1.nrrd

L None
F None
B Normal_1 (252, 293, 168) 458
S Normal_1_1: ● Segment_1

3D Slicer Information

- {Px, Py, Pz}
- Color Density
- Binary Mask

$$M(\{P_x, P_y, P_z\}) = \begin{cases} 1 & \text{if } \{P_x, P_y, P_z\} \in \text{Segment}_1 \\ 0 & \text{otherwise} \end{cases}$$

Extraction Methodologies

		Advantages	Drawbacks
METHOD A	Skeletonization - Scikit Image Reduces the 3D volume to a 1D topological skeleton by iteratively 'peeling' layers of voxels until a single-pixel line remains.	• Fast & Automatic. • Preserves Topology. • Voxel-based.	• Pixel-limited precision. • Mathematically weaker.
METHOD B	Mesh - VMTK Computes a Voronoi diagram on a smooth 3D surface mesh to find the geometric center and the exact maximum inscribed sphere radius.	• Mesh- based. • Mathematically better. • Provides inscribed radius.	• Not automatic. • Requires manual seed points selection.

Hybrid Approach

1. Voxel-Based Topological Thinning – Part of Maren's Work

Used to **separate RCA and LCA arteries** and **discover seed points** (ostium and branch tips) in an automated way.

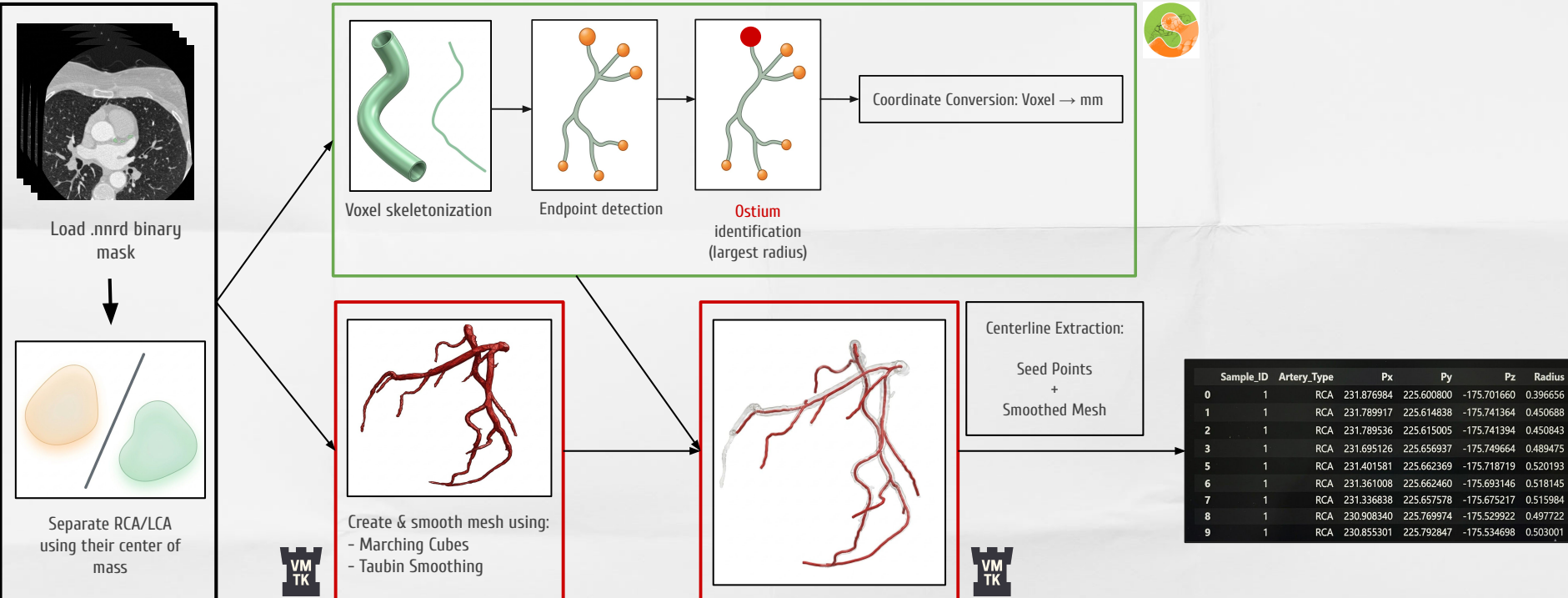
2. VMTK Voronoi-Based Centerline Extraction

Algorithm to compute smooth **centerlines** with maximum inscribed sphere radius.

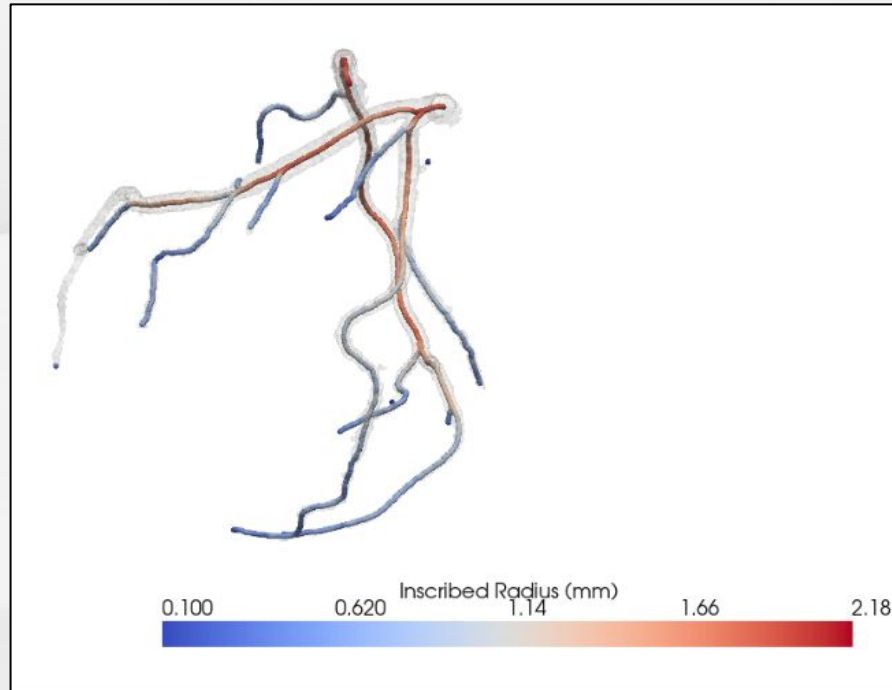
By using skeletonization to automate the seed selection and then feeding those seeds into VMTK, we get the **best of both worlds**:

A fully automated pipeline that outputs centerlines with points and radius data!

Hybrid Approach Architecture



Results: Normal_1.nrrd



Total centerline points : 7293

RCA : 3990 points

LCA : 3303 points

	Sample_ID	Artery_Type	Px	Py	Pz	Radius	
	0	1	RCA	231.876984	225.600800	-175.701660	0.396656
	1	1	RCA	231.789917	225.614838	-175.741364	0.450688
	2	1	RCA	231.789536	225.615005	-175.741394	0.450843
	3	1	RCA	231.695129	225.656937	-175.749664	0.489475
	4	1	RCA	231.635956	225.663040	-175.763458	0.503761
	5	1	RCA	231.401581	225.662369	-175.718719	0.520193
	6	1	RCA	231.361008	225.662460	-175.693146	0.518145
	7	1	RCA	231.336838	225.657578	-175.675217	0.515984
	8	1	RCA	230.908340	225.769974	-175.529922	0.497722
	9	1	RCA	230.855301	225.792847	-175.534698	0.503001
	10	1	RCA	230.549316	225.853973	-175.411545	0.497100
	11	1	RCA	230.402863	225.923401	-175.402832	0.511882
	12	1	RCA	230.398300	225.925766	-175.402039	0.513606
	13	1	RCA	230.288620	225.982697	-175.382935	0.554980
	14	1	RCA	230.236420	225.997314	-175.368713	0.570226
	15	1	RCA	230.076065	226.068695	-175.346786	0.578472
	16	1	RCA	229.832413	226.073135	-175.362579	0.575202
	17	1	RCA	229.785339	226.088486	-175.361145	0.573045
	18	1	RCA	229.729980	226.115524	-175.360565	0.567671
	19	1	RCA	229.522430	226.164352	-175.316956	0.538843



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